LUT School of Energy Systems

Project Work

Technology and Engineering Sciences / LES10A260 Technical Computing Software

Precipitation Forecasting Using Fourier Series Analysis

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ABSTRACT

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This project predicts the total monthly precipitation for April at the Kankaanpää Niinisalo observation station by using historical data. Since daily precipitation data has many zero values and is hard to model directly, the data needs to be summed into monthly totals. The model needs to handle both long-term trends between years and seasonal changes between summer and winter. A Fourier series is used to model the seasonal cycle, and a first-degree polynomial is fitted using the least squares method to find the long-term trend. The models are built in MATLAB, and the final forecast error is estimated using mean squared error.

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Declaration of AI Use

In accordance with LUT University guidelines, the author declares the use of Artificial Intelligence in the preparation of this project work.

Specifically, the Gemini model was used as a planning and coding assistant. It was used to help find relevant sources via Google Scholar and review them to find the best fit for this specific project. It was also used to assist in writing the MATLAB script used for the numerical calculations. Gemini was essentially a private tutor to the author and taught the new methods needed in the execution. All mathematical concepts, data inputs, and the final interpretation of the results were reviewed, directed, and verified by the author.

1 Introduction

Predicting precipitation is important for many areas like agriculture and urban planning (Abdi & Dinpazhoh 2025). In the real world, weather agencies use complex physical models of the atmosphere to predict the weather. But when trying to predict the weather for just one location using only historical data, the problem becomes a time-series extrapolation and curve-fitting task.

The goal of this project is to predict the total precipitation for April at the Kankaanpää Niinisalo weather station. The mathematical model used for this must account for two main things: long-term variation (difference between years) and seasonal variation (winter summer cycle).

Because daily precipitation data jumps around a lot and has many days with zero rain, the raw data is first grouped into monthly sums. To solve the forecasting problem, this work uses a Fourier series to model the seasonal wave and a polynomial fit to find the long-term trend. A Fourier series represents a periodic function using sine and cosine waves, which makes it a good mathematical tool for analyzing seasonal rainfall cycles (Abdi & Dinpazhoh 2025). The model is built using MATLAB and its accuracy is evaluated with mean squared error.

2 Methods

Modeling both yearly and seasonal changes requires two different mathematical approaches. To model the seasonal cycle of precipitation, a Fourier series is used (Afshar & Fahmi 2012). The seasonal wave is modeled with the following function:

$$P(t) = A_0 + \sum_{n=1}^{k} \left(A_n \cos\left(\frac{\pi n t}{T}\right) + B_n \sin\left(\frac{\pi n t}{T}\right) \right) \tag{1}$$

In Equation 1, the coefficients A_0, A_n, and B_n are calculated over the time period T, which is the length of the annual weather cycle (Afshar & Fahmi 2012).

To model the long-term variation between different years, a first-degree polynomial (linear trendline) is used. This polynomial is fitted to the weather data using the method of least squares. This numerical method minimizes the sum of the squared differences between the actual data points and the fitted curve (Kiusalaas 2015). The final forecast is simply the sum of the polynomial trend and the Fourier wave.

The practical computation is executed in MATLAB. Historical daily precipitation data from the Finnish Meteorological Institute (FMI) for the Kankaanpää Niinisalo station is first pre-processed. Any missing daily sensor values are estimated using piecewise linear interpolation, which calculates the mathematical slope between the nearest valid data points to safely bridge gaps. The data is then aggregated into monthly totals to filter out stochastic daily noise.

To compute the linear climate trend, MATLAB's polynomial functions construct a Vandermonde matrix and apply QR decomposition to determine the optimal slope and intercept. For the seasonal wave, rather than relying on automated black-box curve-fitting toolboxes, the Fourier coefficients (A_n and B_n) are computed by manually constructing an overdetermined design matrix. MATLAB's left division operator solves this system by utilizing QR decomposition, factoring the matrix into orthogonal and upper triangular components. This approach ensures a highly stable numerical solution by avoiding the floating-point rounding errors associated with direct matrix inversion. Finally, the goodness-of-fit of the combined model is calculated using mean squared error.

3 Results

The MATLAB implementation successfully processed the historical daily data from the Kankaanpää Niinisalo station, aggregating it into monthly totals and fitting the dual-component mathematical model. The final results of the time-series forecasting are visualized in Figure 1, which displays the actual historical data alongside the fitted mathematical model and the future predicted point.

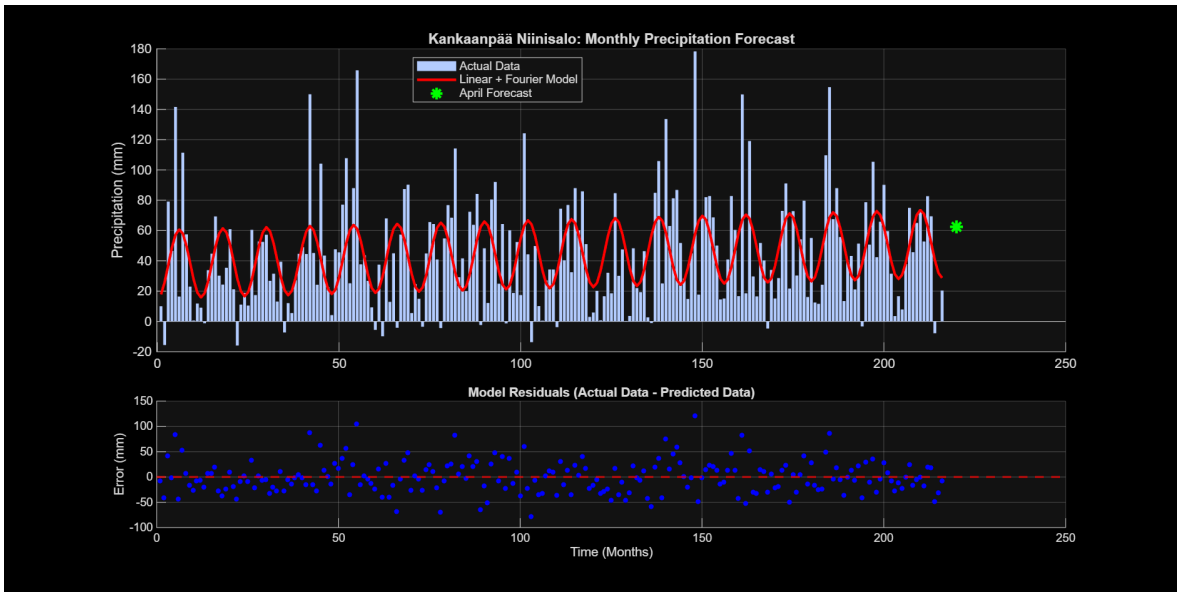


Figure 1: Top: Historical monthly precipitation at Kankaanpää Niinisalo with the fitted Fourier/polynomial model and the forecasted April target. Bottom: The residual plot showing the prediction error for each time step.

Based on the combined first-degree polynomial trendline and the Fourier series seasonal wave, the model predicts that the total cumulative precipitation for the target month of April will be 62.36 mm.

To ensure the most accurate long-term trend was selected, two different polynomial degrees were tested simultaneously. The primary model utilized a 1st-degree (linear) polynomial, resulting in a Mean Squared Error (MSE) of 1068.10. A 2nd-degree (quadratic) polynomial was also fitted to test for non-linear acceleration in the climate trend. However, the quadratic model yielded an MSE of 1066.61. Because the higher-order polynomial did not provide a mathematically significant improvement

to the error rate, the simpler linear trend was retained for the final forecast to avoid over-fitting the data.

4 Discussion

The results demonstrate that a purely mathematical time-series model can successfully capture the broad strokes of local weather patterns. The Fourier series effectively tracked the cyclical wet and dry seasons, while the linear trendline established a stable long-term baseline across the dataset.

However, visual analysis of the residual plot (Figure 1, bottom) highlights the primary limitation of solving this problem using only historical curve-fitting. The mathematical model assumes weather follows a smooth, continuous, and highly regular pattern. In reality, precipitation is highly chaotic. The residuals show massive, seemingly random scatter above and below the zero-error line. A single massive thunderstorm or an unusually dry summer creates extreme spikes and dips in the historical data that a simple continuous wave cannot predict, resulting in the high MSE.

In a real-world scenario, meteorological agencies do not rely exclusively on historical extrapolation. They use complex data assimilation and physical atmospheric models that track actual current conditions (like air pressure, wind, and humidity) to generate near-term forecasts. While this project's mathematical approach provides a solid baseline expectation for April (62.36 mm) based on past averages, it cannot account for sudden, chaotic weather events. Nevertheless, the method provides a useful analytical tool that could be practically applied in fields like agriculture or urban infrastructure planning, where understanding the general seasonal expectation over decades is more critical than predicting exact daily rainfall.

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